



```
In [16... import warnings
warnings.filterwarnings('ignore')
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
```

```
In [15... df = pd.read_csv('/content/Airbnb_data.csv', skipinitialspace = True )
df.head()
```

Out[15]:

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	nei
0	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	
1	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	
2	1002403	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	
3	1002755	NaN	85098326012	unconfirmed	Garry	Brooklyn	
4	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	

5 rows × 26 columns

```
In [17... df.columns
```

```
Out[17]: Index(['id', 'NAME', 'host id', 'host_identity_verified', 'host name',
              'neighbourhood group', 'neighbourhood', 'lat', 'long', 'countr
              y',
              'country code', 'instant_bookable', 'cancellation_policy', 'room
              type',
              'Construction year', 'price', 'service fee', 'minimum nights',
              'number of reviews', 'last review', 'reviews per month',
              'review rate number', 'calculated host listings count',
              'availability 365', 'house_rules', 'license'],
              dtype='object')
```

Checking missing values

In [20... `print(df.isnull().sum())`

id	0
NAME	250
host id	0
host_identity_verified	289
host name	406
neighbourhood group	29
neighbourhood	16
lat	8
long	8
country	532
country code	131
instant_bookable	105
cancellation_policy	76
room type	0
Construction year	214
price	247
service fee	273
minimum nights	409
number of reviews	183
last review	15893
reviews per month	15879
review rate number	326
calculated host listings count	319
availability 365	448
house_rules	52131
license	102597
dtype: int64	

In [22... `df.info()`

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 102599 entries, 0 to 102598
Data columns (total 26 columns):
#   Column                                Non-Null Count  Dtype
---  ---                                -
0   id                                    102599 non-null  int64
1   NAME                                102349 non-null  object
2   host id                             102599 non-null  int64
3   host_identity_verified              102310 non-null  object
4   host name                           102193 non-null  object
5   neighbourhood group                 102570 non-null  object
6   neighbourhood                       102583 non-null  object
7   lat                                 102591 non-null  float64
8   long                                102591 non-null  float64
9   country                             102067 non-null  object
10  country code                        102468 non-null  object
11  instant_bookable                    102494 non-null  object
12  cancellation_policy                 102523 non-null  object
13  room type                           102599 non-null  object
14  Construction year                   102385 non-null  float64
15  price                               102352 non-null  object
16  service fee                         102326 non-null  object
17  minimum nights                      102190 non-null  float64
18  number of reviews                   102416 non-null  float64
19  last review                         86706 non-null  object
20  reviews per month                   86720 non-null  float64
21  review rate number                  102273 non-null  float64
22  calculated host listings count      102280 non-null  float64
23  availability 365                    102151 non-null  float64
24  house_rules                         50468 non-null  object
25  license                             2 non-null      object
dtypes: float64(9), int64(2), object(15)
memory usage: 20.4+ MB

```

Handling Missing Values

```
In [26... df['last review'] = pd.to_datetime(df['last review'], errors = 'coerce')
```

```
In [29... df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 102599 entries, 0 to 102598
Data columns (total 26 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   id                                    102599 non-null  int64
1   NAME                                102349 non-null  object
2   host id                             102599 non-null  int64
3   host_identity_verified              102310 non-null  object
4   host name                           102193 non-null  object
5   neighbourhood group                 102570 non-null  object
6   neighbourhood                       102583 non-null  object
7   lat                                 102591 non-null  float64
8   long                                102591 non-null  float64
9   country                             102067 non-null  object
10  country code                        102468 non-null  object
11  instant_bookable                    102494 non-null  object
12  cancellation_policy                 102523 non-null  object
13  room type                           102599 non-null  object
14  Construction year                   102385 non-null  float64
15  price                               102352 non-null  object
16  service fee                         102326 non-null  object
17  minimum nights                      102190 non-null  float64
18  number of reviews                   102416 non-null  float64
19  last review                         52371 non-null   datetime64[ns]
20  reviews per month                   86720 non-null   float64
21  review rate number                  102273 non-null   float64
22  calculated host listings count      102280 non-null   float64
23  availability 365                    102151 non-null   float64
24  house_rules                         50468 non-null   object
25  license                             2 non-null       object
dtypes: datetime64[ns](1), float64(9), int64(2), object(14)
memory usage: 20.4+ MB

```

```
In [30...] df.fillna({'reviews per month':0, 'last review': df['last review'].min()})
```

```
In [31...] df.dropna(subset= ['NAME', 'host name'], inplace = True)
```

```
In [32...] print(df.isnull().sum())
```

id	0
NAME	0
host id	0
host_identity_verified	276
host name	0
neighbourhood group	26
neighbourhood	16
lat	8
long	8
country	526
country code	122
instant_bookable	96
cancellation_policy	70
room type	0
Construction year	200
price	239
service fee	268
minimum nights	403
number of reviews	182
last review	0
reviews per month	0
review rate number	314
calculated host listings count	318
availability 365	420
house_rules	51867
license	101947
dtype:	int64

```
In [35... df = df.drop(columns = ['license', 'house_rules'], errors = 'ignore')
```

```
In [37... df.head()
```

Out[37]:

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	nei
0	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	
1	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	
2	1002403	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	
4	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	
5	1004098	Large Cozy 1 BR Apartment In Midtown East	45498551794	verified	Michelle	Manhattan	

5 rows × 24 columns

```
In [38... df['price'] = df['price'].replace('[\$,]', '', regex = True).astype(float)
df['service fee'] = df['service fee'].replace('[\$,]', '', regex= True).ast
```

```
In [39... df.head()
```

Out[39]:

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	nei
0	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	
1	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	
2	1002403	THE VILLAGE OF HARLEM....NEW YORK !	78829239556	NaN	Elise	Manhattan	
4	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	
5	1004098	Large Cozy 1 BR Apartment In Midtown East	45498551794	verified	Michelle	Manhattan	

5 rows × 24 columns

Remove Duplicates

```
In [40...] df.drop_duplicates(inplace = True)

In [41...] df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
Index: 101410 entries, 0 to 102057
Data columns (total 24 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   id                                           101410 non-null  int64
1   NAME                                         101410 non-null  object
2   host id                                     101410 non-null  int64
3   host_identity_verified                     101134 non-null  object
4   host name                                  101410 non-null  object
5   neighbourhood group                        101384 non-null  object
6   neighbourhood                              101394 non-null  object
7   lat                                         101402 non-null  float64
8   long                                        101402 non-null  float64
9   country                                     100884 non-null  object
10  country code                               101288 non-null  object
11  instant_bookable                           101314 non-null  object
12  cancellation_policy                         101340 non-null  object
13  room type                                  101410 non-null  object
14  Construction year                           101210 non-null  float64
15  price                                        101171 non-null  float64
16  service fee                                101142 non-null  float64
17  minimum nights                             101016 non-null  float64
18  number of reviews                          101228 non-null  float64
19  last review                                101410 non-null  datetime64[ns]
20  reviews per month                          101410 non-null  float64
21  review rate number                         101103 non-null  float64
22  calculated host listings count              101092 non-null  float64
23  availability 365                           100990 non-null  float64
dtypes: datetime64[ns](1), float64(11), int64(2), object(10)
memory usage: 19.3+ MB

```

Descriptive Statistics

In [42... `df.describe()`

Out[42]:

	id	host id	lat	long	Construction year	price
count	1.014100e+05	1.014100e+05	101402.000000	101402.000000	101210.000000	101171.000000
mean	2.920959e+07	4.926155e+10	40.728082	-73.949663	2012.486908	625.381000
min	1.001254e+06	1.236005e+08	40.499790	-74.249840	2003.000000	50.000000
25%	1.507574e+07	2.459183e+10	40.688730	-73.982570	2007.000000	340.000000
50%	2.922911e+07	4.912069e+10	40.722300	-73.954440	2012.000000	625.000000
75%	4.328308e+07	7.399747e+10	40.762750	-73.932340	2017.000000	913.000000
max	5.736742e+07	9.876313e+10	40.916970	-73.705220	2022.000000	1200.000000
std	1.626820e+07	2.853703e+10	0.055850	0.049474	5.765130	331.609000

Visualisation

Distribution of listing prices:

In [44...]

```
plt.figure(figsize=[10,6])
sns.histplot(df['price'], bins = 50, kde = True, color= 'red')
plt.title('Distribution of listing price')
plt.xlabel('Price $')
plt.ylabel('Frequency')
plt.show()
```



The histogram shows a fairly even distribution of listing prices across different price ranges, indicating no particular concentration of listings in any specific price range. The kde line helps visualize this spread even more clearly, confirming that the dataset contains listings with a wide variety of prices

Distribution of different room types

```
In [45... df['room type']
```

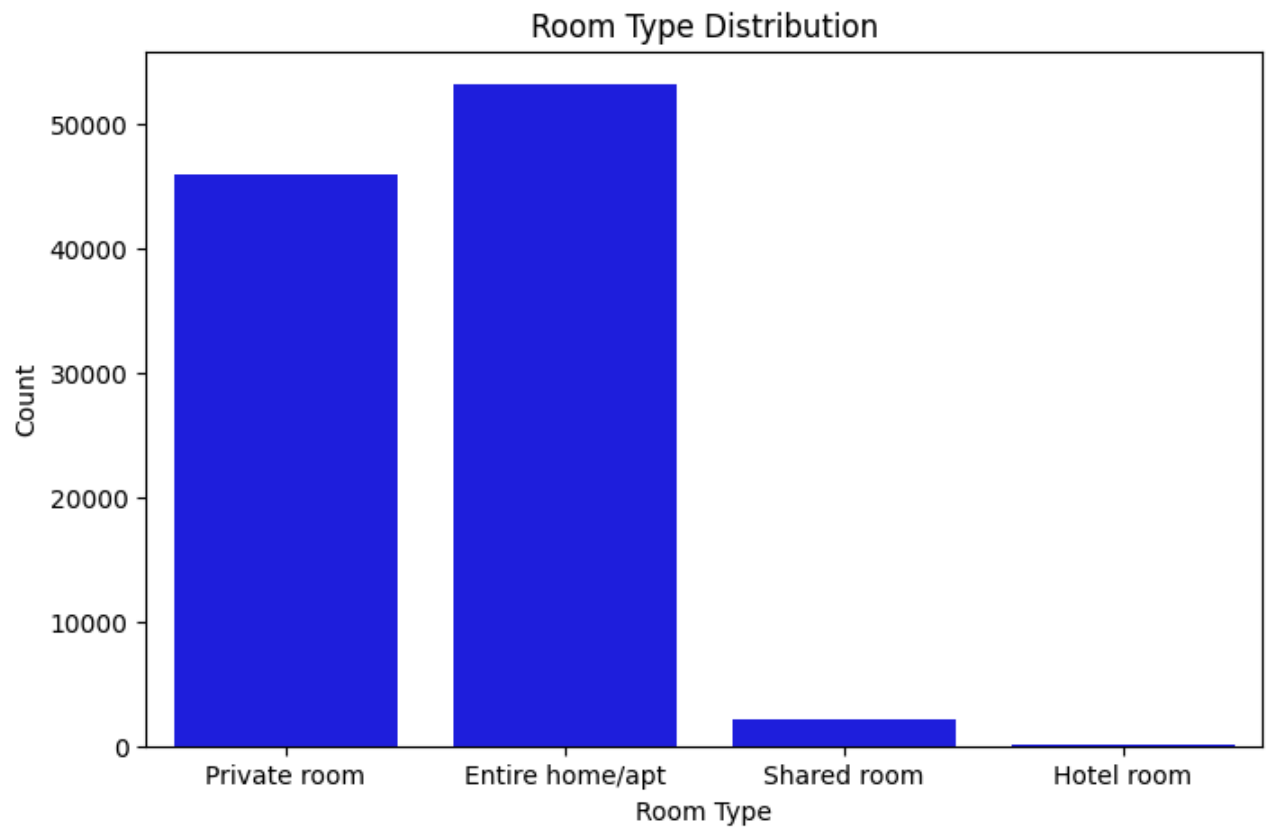
Out[45]:

room type	
0	Private room
1	Entire home/apt
2	Private room
4	Entire home/apt
5	Entire home/apt
...	...
102053	Private room
102054	Private room
102055	Entire home/apt
102056	Private room
102057	Entire home/apt

101410 rows × 1 columns

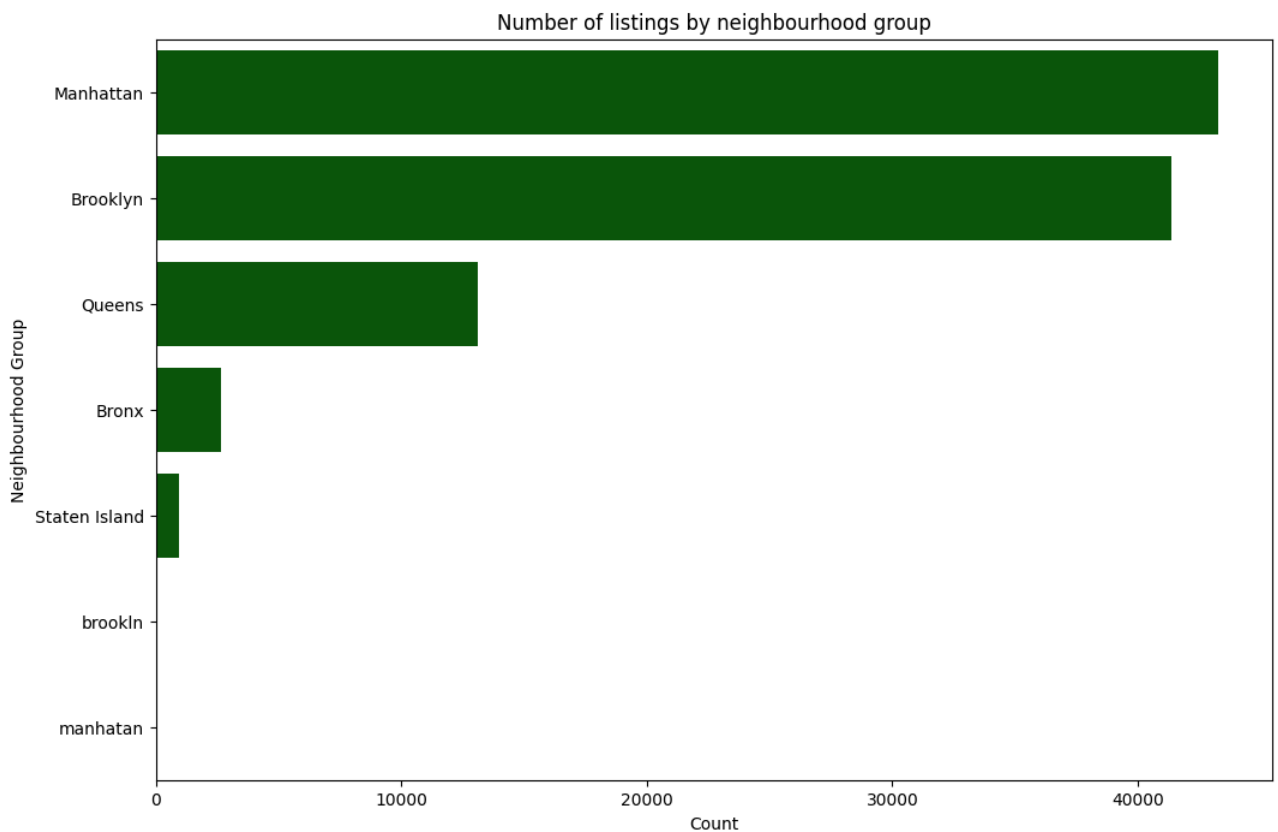
dtype: object

```
In [48... plt.figure(figsize= (8,5))
sns.countplot(x = 'room type', data = df, color = 'blue')
plt.title('Room Type Distribution')
plt.xlabel('Room Type')
plt.ylabel('Count')
plt.show()
```



Distribution of listings across different neighbourhoods

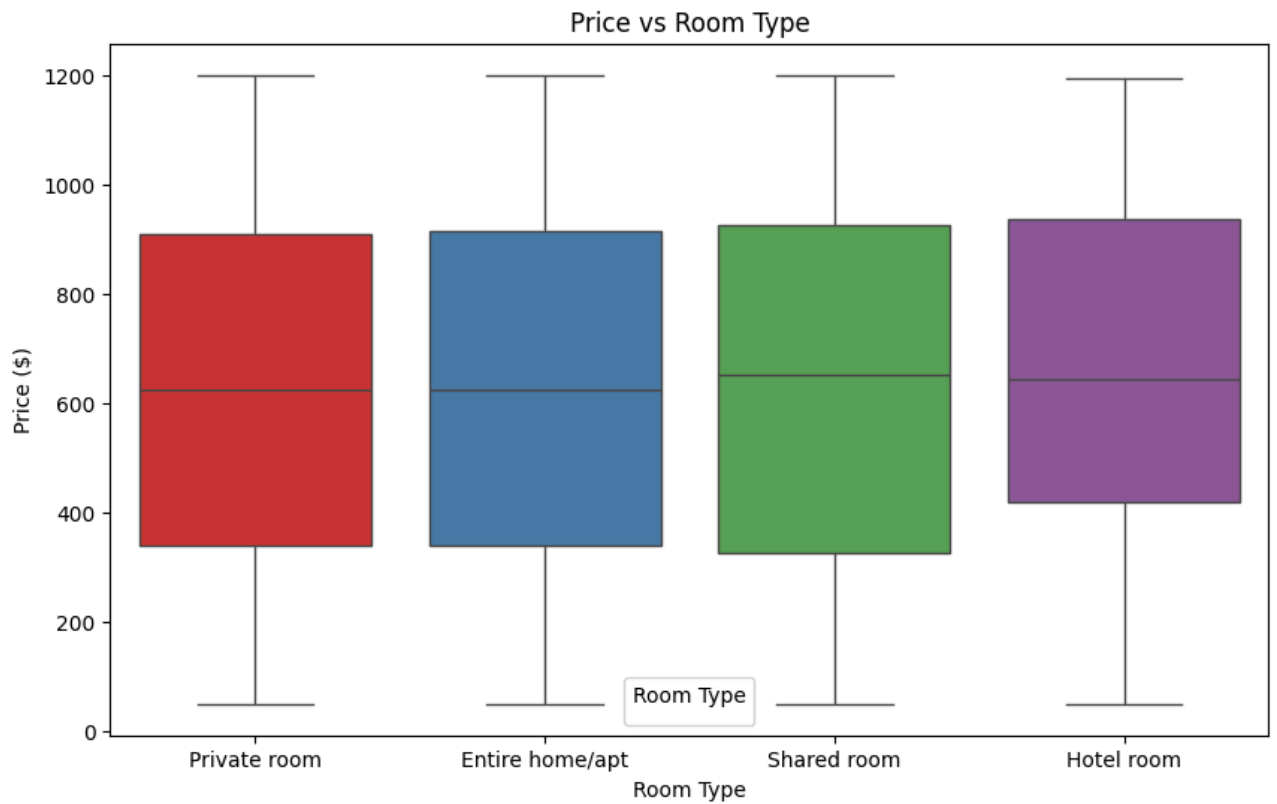
```
In [50... plt.figure(figsize = (12,8))
sns.countplot(y = 'neighbourhood group', data = df, color = 'darkgreen',
plt.title('Number of listings by neighbourhood group')
plt.xlabel('Count')
plt.ylabel('Neighbourhood Group')
plt.show()
```



Manhattan, Brooklyn and Queens are the neighbourhood groups where most of the properties are listed

Relationship between Price and Room Type

```
In [51... plt.figure(figsize = (10,6))
sns.boxplot(x = 'room type', y = 'price', hue = 'room type', data = df, p
plt.title('Price vs Room Type')
plt.xlabel('Room Type')
plt.ylabel('Price ($)')
plt.legend(title = 'Room Type')
plt.show()
```



Change of number of reviews over time

```
In [52... df['last review'] = pd.to_datetime(df['last review'])
reviews_over_time = df.groupby(df['last review'].dt.to_period('M')).size()

plt.figure(figsize = (12,6))
reviews_over_time.plot(kind = 'line',color = 'red')
plt.title('Number of Reviews over time')
plt.xlabel('Date')
plt.ylabel('Number of Reviews')
plt.show()
```

